

At the top right is a firework `SpriteSheetAnimation` object, which comes from a single image containing 35 subimages. This animation is set to loop, so you see it run continuously.

At the bottom right is a walking `SpriteSheetAnimation` from a single image with 36 subimages. When you click Start, the animation plays through all the images once.

The full source code of this program is available in the *AnimationExample/Main_AnimationExample.py* file along with the rest of the book's resources.

The program instantiates two `Animation` objects (the small dinosaur and the T-rex) and two `SpriteSheetAnimation` objects (the firework and the walking person). When a button below the little dinosaur is clicked, we call the appropriate method of the dinosaur animation object. A click on the little dinosaur or T-rex results in a call to the `start()` method of that animation.

The program shows that multiple animations can run at the same time. This works because the main loop calls the `update()` and `draw()` methods of *each* animation in every frame in the main loop, and each animation makes its own decision about keeping the current image or showing the next image.

Summary

In this chapter, we explored the mechanisms required in an animation class by building our own `SimpleAnimation` and `SimpleSpriteSheetAnimation` classes. The former is made up of multiple images, whereas the latter uses a single larger image that contains multiple subimages.

These two classes have different initializations, but the remaining methods of the classes are identical. I explained the process of merging the two classes by building a common abstract base class.

I then introduced the `Animation` class and the `SpriteSheetAnimation` class in `pygwidgets`. I explained that these two classes only implement their own versions of the `__init__()` method, inheriting their other methods from the common base class, `PygAnimation`. I concluded by showing a demonstration program that provides examples of animations and sprite sheet animations.